SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA51 COURSE NAME: Cryptography and Network Security

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COURSE OUTCOME COVERED

CO3: *Examine cryptographic hash algorithms, digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication, and non-repudiation.CO* (BL4 , BL5)

Assessment Tool Weightage: 10%

QUESTIONS: 5 **TOTAL MARKS: 50**

Annexure A
MCQ

Code of Conduct:

I P.S.Surabhi Surya(192421267)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Surabhi

Annexure A
MCQ

MCQ (Additional Set)

1. Which property ensures the same input always produces the same hash?
   A) Avalanche effect
   B) Determinism
   C) Collision resistance
   D) Diffusion
2. What is the primary purpose of SHA-3?
   A) Faster encryption
   B) Alternative secure hashing standard
   C) Data compression
   D) Key exchange
3. Which attack tries all possible inputs to match a hash?
   A) Replay attack
   B) Brute-force attack
   C) MITM attack
   D) DoS attack
4. What is a salted hash?
   A) Encrypted hash
   B) Hash with random data added
   C) Shortened hash
   D) Reversed hash
5. Which algorithm is widely used for password hashing?
   A) AES
   B) RSA
   C) bcrypt
   D) DES
6. What does second pre-image resistance ensure?
   A) Same input produces same output
   B) Hard to find different input with same hash
   C) Fast hashing
   D) Small hash size
7. What is the output size of SHA-512?
   A) 256 bits
   B) 128 bits
   C) 512 bits
   D) 1024 bits
8. Which technique slows down brute-force attacks?
   A) Encryption
   B) Key exchange
   C) Key stretching
   D) Compression
9. What is the purpose of a nonce?
   A) Encryption
   B) Prevent replay attacks
   C) Compress data
   D) Generate keys
10. Which function is irreversible?
   A) Encryption
   B) Hashing

C) Compression
D) Encoding

11. Which algorithm is used in DSS?
    A) AES
    B) DSA
    C) DES
    D) RC4
12. What is required to verify a digital signature?
    A) Private key
    B) Public key
    C) Session key
    D) Shared key
13. What does a digital signature guarantee?
    A) Confidentiality
    B) Integrity and authenticity
    C) Compression
    D) Speed
14. What happens if message digest changes?
    A) Signature remains valid
    B) Verification fails
    C) Encryption fails
    D) Speed increases
15. Which attack involves forging a signature?
    A) Replay
    B) Forgery
    C) DoS
    D) Sniffing
16. What is the role of a hash in signatures?
    A) Encrypt data
    B) Reduce size
    C) Ensure integrity
    D) Store keys
17. Which key is kept secret in signatures?
    A) Public key
    B) Private key
    C) Session key
    D) Shared key
18. What ensures trust in certificates?
    A) Encryption
    B) Certificate Authority
    C) Compression
    D) Hash size
19. What is certificate revocation?
    A) Key generation
    B) Canceling validity
    C) Encryption
    D) Compression
20. What is OCSP used for?
    A) Encryption
    B) Certificate validation
    C) Compression
    D) Routing
21. What is the main goal of Kerberos tickets?
    A) Encryption
    B) Authentication

C) Compression
D) Routing

22. Which protocol uses tickets for authentication?
A) SSL
B) Kerberos
C) FTP
D) HTTP

23. What is a session key?
A) Permanent key
B) Temporary key
C) Public key
D) Private key

24. What is mutual authentication?
A) One-way authentication
B) Both parties verify each other
C) Anonymous login
D) Weak authentication

25. What ensures freshness in Kerberos?
A) Encryption
B) Timestamp
C) Compression
D) Hashing

26. What is a replay attack?
A) Data modification
B) Resending valid data
C) Encryption
D) Compression

27. What is the function of an authenticator in Kerberos?
A) Data storage
B) Identity proof
C) Encryption
D) Compression

28. What happens if system clocks differ in Kerberos?
A) Faster authentication
B) Failure
C) Better security
D) No effect

29. What is the KDC composed of?
A) Client & Server
B) AS & TGS
C) Firewall & IDS
D) Router & Switch

30. What is the AS in Kerberos?
A) Application Server
B) Authentication Server
C) Access Server
D) Admin Server

31. What is X.509 certificate format used in?
A) Networking
B) Digital certificates
C) Compression
D) Routing

32. What identifies the certificate owner?
A) Public key
B) Subject field

C) Signature
D) Hash

33. What ensures certificate integrity?
A) Encryption
B) Digital signature
C) Compression
D) Routing

34. What is a root certificate?
A) Expired certificate
B) Self-signed trusted certificate
C) Invalid certificate
D) Temporary certificate

35. What is certificate pinning?
A) Storing keys
B) Binding certificate to host
C) Encryption
D) Compression

36. What is the risk of weak CA validation?
A) Faster speed
B) MITM attacks
C) Compression
D) Routing

37. What is the purpose of trust chain?
A) Encryption
B) Validate certificate hierarchy
C) Compress data
D) Routing

38. What is digital certificate used for?
A) Compression
B) Authentication
C) Routing
D) Storage

39. What ensures non-repudiation?
A) Encryption
B) Digital signature
C) Compression
D) Routing

40. What is a hash collision?
A) Same input
B) Same output for different inputs
C) Encryption
D) Compression

41. Which hash is strongest?
A) MD5
B) SHA-1
C) SHA-256
D) CRC

42. What is a one-way function?
A) Reversible
B) Irreversible
C) Fast
D) Small

43. What ensures authentication in PKI?
A) Encryption
B) Certificates

C) Compression
D) Routing

44. What is key escrow?
A) Key deletion
B) Key storage by third party
C) Encryption
D) Compression

45. What is identity spoofing?
A) Encryption
B) Fake identity
C) Compression
D) Routing

46. What is trust anchor?
A) Root CA
B) Client
C) Server
D) Router

47. What ensures integrity of message?
A) Encryption
B) Hash
C) Compression
D) Routing

48. What is digital envelope?
A) Hash
B) Encrypted message + key
C) Compression
D) Routing

49. What is certificate fingerprint?
A) Encryption
B) Hash of certificate
C) Compression
D) Routing

50. What is PKI used for?
A) Routing
B) Secure communication
C) Compression
D) Storage